MIL-P-14273 (Ord)
23 March 1956
SUPERSEDING
MPD-537A
7 April 1955

MILITARY SPECIFICATION

POD KIT, RADAR REFLECTIVE,
AERIAL TARGET M47

1. SCOPE

1.1 This specification covers the requirements for the M47 Aerial Target Radar Reflective Pod Kit used to improve the radar reflectivity of the OQ-19B Aerial Target Drone.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standard, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal
UU-T-116 - Tape, Paper, Gummed, Water-Resistant

Military
JAN-P-100 - Packaging and Packing for Overseas Shipment, General Specification
MIL-P-116 - Preservation, Methods of

All specifications referenced on drawings specified below.

FSC 5840

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STANDARD

Military
MIL-STD-171 - Preparation, Painting and Finishing, Metal and Wood Surfaces

DRAWINGS

Ordnance Corps
8188773 - Packing and Crating For Radar Reflector (Assembly)
8188799 - Radar Reflector (Assembly and Installation)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. - Materials shall conform to the requirements of this specification, with the applicable drawings and material specifications. Deviations from specified material shall only be made upon written approval by the contracting officer. Materials designated on the drawings as "commercial" shall be of the best quality available, suitable for the intended purpose, and free from defects which may affect their serviceability or appearance.

3.2 Assembly. - The pod kit shall conform to the requirements of Ordnance Corps Drawing 8188799 and shall consist of the following:

a. Ten (10) pairs of retroreflector pods.
b. Thirty (30) pairs of wing mounting brackets and associated hardware.

3.2.1 Cluster corners.- The arrangement of the corners in the cluster shall be such as to give the most complete coverage possible. The corners are to be usable at radar frequencies from 2,700 mc to 30,000 mc.

3.3 Aerodynamic and structural.

3.3.1 Configuration. - The external configuration of the pod shall be arranged so as to provide the best radar reflectivity characteristics possible. The fineness ratio of the configuration shall not be less than four.

3.3.2 Durability. - The pods, when mounted on the slow speed aerial target drone, shall be capable of withstanding the effects of a true air speed of 220 knots at all altitudes up to 25,000 feet without structual deformation to the reflector cluster or to the individual corners to such a

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degree as to affect the reflectivity characteristics of the pod. The pods shall be capable of withstanding the normal air and shock loads associated with the recovery sequence of the target.

3.4 Electronics.

3.4.1 Surface deviations. - No surface of any corner reflector shall be allowed to deviate from its true position with respect to all other surfaces by more than 0.250 inch displacement at the edge. These same surfaces shall not deviate from being plane surfaces by more than 0.125 inch maximum displacement measured perpendicular to the surface.

3.4.2 Reflectivity. - The basic power reflectivity of the material from which the corners are fabricated shall not be less than 85 percent of the reflectivity of a perfect reflector at X-Band. This reflectivity must be maintained for long storage periods.

3.4.3 Transmission coefficient. - The radome material shall have a transmission coefficient of at least 85 percent in power at X-Band and this shall not change by being soaked in water and then air dried.

3.5 Protective requirements. - Unless otherwise specified on the applicable drawings and specifications, protective requirements shall be as specified in Standard MIL-STD-171.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. - Unless otherwise specified by the contracting officer, sampling shall be 100 percent.

4.2 Inspection. - Each pod kit shall be given a careful dimensional and visual inspection to determine compliance with the requirements of this specification and applicable drawings and specifications. Noncompliance with any of the requirements specified or referenced herein shall be cause for rejection subject to the resubmission requirements.

4.3 Resubmission. - Pod kits which have been rejected may be reworked or parts replaced to correct the defect and resubmitted for acceptance. Prior to resubmitting, full particulars concerning previous rejection and action taken to correct the defects shall be furnished the inspector. Resubmitted assemblies shall be reinspected in accordance with 4.2. Unless otherwise specified by the procuring activity, disposition of reinspection rejects shall be at the discretion of the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 Cleaning. - Cleaning shall be in accordance with Method C-1, Specification MIL-P-116.

5.2 Drying. - Unless otherwise specified in the contract or order, the contractor may use any drying procedure specified in MIL-P-116.

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5.3 Preservation. - Unless otherwise specified in the contract or order, the selection of a preservative, where required, from Table I of Specification MIL-P-116 shall be at the option of the contractor.

5.4 Packaging.

5.4.1 Level A - Military package. - The unit package shall be in accordance with Ordnance Corps Drawing 8188773. Attaching hardware shall be packed in accordance with Ordnance Corps Drawing 8188773. Closures for containers shall conform to Specification UU-T-116.

5.4.2 Level B - Limited Military package. - Same as 5.4.1.

5.4.3 Level C - Minimum military package. - Not applicable.

5.5 Packing.

5.5.1 Level A - Military pack. - Exterior pack shall be in accordance with Ordnance Corps Drawing 8188773.

5.5.2 Level B - Limited military pack. - Same as 5.5.1.

5.5.3 Level C - Minimum military pack. - Not applicable.

5.5.4 Completed pack. - Completed pack shall be capable of passing the applicable test specified in Specification JAN-P-100.

5.6 Markings. - In addition to any special marking, as required by contract or order, interior packages and exterior shipping containers shall be marked in accordance with Ordnance Corps Drawing 8188773.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Level of protection required. (See 5.4)

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